

Metadata Documentation & Compliance Note

Project Title

Mental Health Services Data Set (MHSDS) – Combined Metadata Documentation

Document Version

Version: 1.0

Prepared Date: 08 May 2026

Prepared For: AI Mental Health Assistance Research Project

1. Overview

This metadata document has been prepared using the Excel workbook:

File Name: mhsds-combined-metadata-13032026.xlsx.xlsx

The workbook contains structured metadata, validation rules, integrity checks, annual bulletin metadata, and mental health reporting definitions related to Mental Health Services Data Set (MHSDS) reporting.

The purpose of this document is to:

- Describe the structure and meaning of the dataset metadata.
 - Define reporting standards and validation rules.
 - Provide governance and compliance documentation.
 - Support AI-based mental health analytics and reporting systems.
-

2. Workbook Structure Summary

Sheet Name	Purpose
Front Sheet	Introduction and dataset overview
Version Control	Change tracking and version history
Metadata - Field List	Definitions and descriptions of dataset fields
Mental Health Monthly Metadata	Monthly reporting metrics and metadata
Annual Bulletin Metadata	Annual publication metrics

Sheet Name	Purpose
MHA Annual Metadata	Mental Health Act annual metrics
Validation - Rules	Data validation conditions and quality rules
Integrity - Rules	Data integrity and consistency checks

3. Metadata Field Documentation

The worksheet **"Metadata - Field List"** contains core metadata definitions used across the reporting framework.

Key Metadata Fields

Metadata Field	Description
Metric ID	Unique identifier assigned to each metric
Metric Name	Official metric title used in reporting
Metric Description	Explanation of the metric calculation or purpose
Reporting Period	Defines whether the metric is monthly, quarterly, or annual
Metric Type	Indicates count, rate, proportion, mean, or median
Count Of	Defines the entity being counted (people, referrals, spells, etc.)
Numerator	Numerator component for calculated metrics
Denominator	Denominator component for rates and proportions
Publication	Source publication or reporting framework
Breakdown	Category used for grouping or segmentation
Geography	Geographic scope of reporting
Data Source	Source system or reporting feed
Update Frequency	Defines how often the metric is refreshed

4. Dataset Characteristics

Data Domain

Mental Health Services and Healthcare Analytics

Data Type

Structured tabular metadata and reporting definitions

Primary Use Cases

- Mental health reporting
- AI mental health analytics
- Healthcare dashboards
- Research publication support
- NHS-style reporting frameworks
- Predictive analytics and statistical modelling

Expected Consumers

- Researchers
 - Healthcare analysts
 - Data scientists
 - Compliance teams
 - Mental health administrators
 - AI model developers
-

5. Validation Rules Summary

The workbook contains a dedicated sheet named **“Validation - Rules”** defining business and data quality validations.

Validation Objectives

- Prevent missing or invalid values
- Ensure metric consistency
- Validate reporting periods
- Maintain formatting standards
- Ensure compliance with publication requirements

Common Validation Categories

Validation Type	Description
Null Checks	Detect missing mandatory values
Format Validation	Ensure IDs and codes follow expected patterns
Range Validation	Ensure values remain within acceptable ranges
Duplicate Detection	Prevent repeated metric entries
Consistency Validation	Cross-check related metric values
Reporting Validation	Ensure correct publication mapping

6. Integrity Rules Summary

The worksheet “**Integrity - Rules**” defines logical consistency requirements.

Integrity Objectives

- Preserve reporting accuracy
- Maintain referential consistency
- Prevent contradictory metrics
- Ensure reliable AI training inputs

Integrity Rule Examples

Rule Category	Purpose
Cross-Field Consistency	Ensures related values align correctly
Metric Dependency Validation	Verifies dependent calculations
Temporal Integrity	Ensures chronological consistency
Publication Integrity	Maintains publication-specific rules
Identifier Integrity	Ensures uniqueness of metric identifiers

7. AI & Analytics Integration Metadata

AI Readiness Assessment

Area	Status
Structured Metadata	Available
Validation Rules	Available
Data Integrity Rules	Available
Reporting Definitions	Available
Standardized Metrics	Available
Governance Support	Available

Suitable AI Applications

- Mental health prevalence prediction
 - Trend analysis
 - Healthcare reporting automation
 - NLP-based healthcare analysis
 - Statistical forecasting
 - AI compliance monitoring
 - Data quality anomaly detection
-

8. Security & Governance Metadata

Governance Area	Description
Data Governance	Controlled metadata definitions and reporting structure
Access Management	Role-based access recommended
Auditability	Validation and integrity rules support audit trails
Data Quality	Multi-layer validation framework available
Standardization	Uniform metric definitions maintained
Traceability	Metric-level traceability supported

9. Compliance Note

Compliance & Regulatory Statement

This metadata framework has been reviewed for compatibility with healthcare analytics governance practices and AI-supported reporting systems.

The workbook structure demonstrates implementation of:

- Metadata governance
 - Validation control mechanisms
 - Reporting consistency standards
 - Data integrity verification
 - Audit-supporting structures
-

Data Privacy & Confidentiality

The metadata workbook primarily contains:

- Reporting definitions
- Metric descriptions
- Validation logic
- Structural metadata

No direct personally identifiable information (PII) was identified in the reviewed metadata sheets.

However, any linked operational datasets used alongside this metadata framework should follow:

- GDPR principles
 - HIPAA-aligned healthcare data protections
 - NHS information governance practices
 - Data minimization principles
 - Access control policies
-

AI Compliance Considerations

For AI or machine learning implementation using this metadata framework:

Recommended Controls

- Bias monitoring during model training
- Validation of derived healthcare predictions
- Human review for critical decisions

- Explainable AI techniques
- Secure logging and audit trails
- Periodic model performance evaluation

Ethical AI Requirements

- Avoid discriminatory predictions
- Maintain fairness across demographic groups
- Use anonymized or pseudonymized data where applicable
- Ensure transparency in automated decision systems

Data Quality Compliance

The inclusion of validation and integrity rule sheets supports:

- Data quality assurance
- Consistency verification
- Regulatory audit readiness
- Reporting standardization
- Statistical reliability

This improves trustworthiness for:

- Healthcare reporting
- Research publications
- AI analytics pipelines
- Government or institutional submissions

Risk Assessment Summary

Risk Area	Risk Level	Mitigation
Missing Metadata	Low	Validation rules implemented
Data Integrity Failure	Low	Integrity rule checks available
AI Bias Risk	Medium	Requires fairness monitoring
Unauthorized Access	Medium	RBAC and encryption recommended
Reporting Inconsistency	Low	Standardized metric framework

10. Recommendations

Technical Recommendations

- Maintain version-controlled metadata updates
- Automate validation checks during ingestion
- Integrate metadata cataloguing tools
- Use centralized governance dashboards
- Maintain schema documentation

AI Recommendations

- Use metadata-driven feature engineering
- Apply explainable AI frameworks
- Monitor model drift continuously
- Conduct periodic compliance audits

Governance Recommendations

- Maintain audit logs
 - Review validation rules quarterly
 - Establish metadata stewardship roles
 - Define formal approval workflows
-

11. Conclusion

The uploaded MHSDS metadata workbook provides a structured and governance-oriented framework for mental health reporting and analytics.

The presence of:

- standardized metadata definitions,
- validation controls,
- integrity rules,
- reporting structures, and
- version control mechanisms

makes the dataset suitable for:

- healthcare analytics,
- AI/ML integration,
- statistical modelling,
- compliance-driven reporting, and
- enterprise data governance applications.

The framework demonstrates strong alignment with modern healthcare data management and AI governance principles.